

SAiDL Assignment 2026

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27th March 2026 Onwards

Core-ML Task

1 Intro

This is my report on the Core-ML task of Attention-based Transformer, trained on WikiText-2. I'll be updating my progress here.

2 Basic Transformer Baseline

From the 'Attention is All You Need' article, the core of the Transformer is the Scaled Dot-Product Attention:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (1)$$

where:

- $Q \rightarrow$ Queries
- $K \rightarrow$ Keys
- $V \rightarrow$ Values
- $d_k \rightarrow$ Dimension of the keys